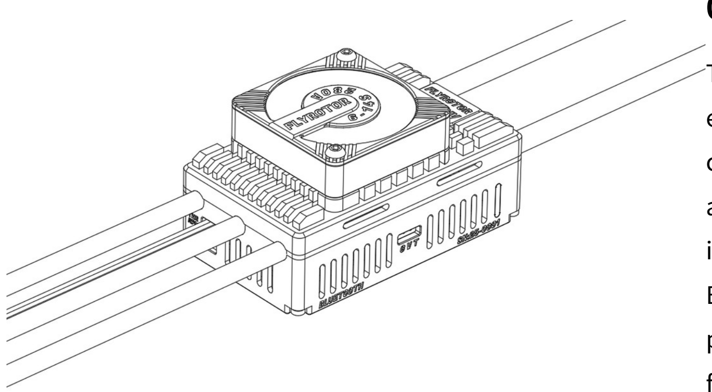
FLYROTOR 280A HV Brushless ESC Manual



01 Disclaimer

Thank you for purchasing the FLYROTOR 280A HV brushless electronic speed controller (ESC). Please read this disclaimer carefully before use. By using this product, you are deemed to have acknowledged and accepted all contents of this disclaimer. Please install and use this product strictly in accordance with this manual. Brushless power systems are powerful, and improper use may cause personal injury or equipment damage. We assume no liability arising from the use of this product or from unauthorized modification of the product, including but not limited to liability for incidental or consequential damages. We reserve the right to change the product design, appearance, performance, and usage requirements without prior notice.

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02 Precautions

① Before using this product, carefully read the documentation for the supporting equipment and aircraft, and make sure the power system configuration is appropriate. Improper matching may overload the motor and damage the ESC.
② Installation involves soldering, wiring, and related operations. Make sure all wires and connection points are properly insulated. Any short circuit may damage the equipment. When soldering product wires, use a soldering tool with suitable power to ensure a firm and reliable connection. Unstable connections may cause abnormal aircraft control, equipment damage, or other unexpected situations.
③ Use this product in a safe environment, away from obstacles, crowds, high-voltage lines, and other potential hazards. Operate strictly within the working conditions specified in this manual, including voltage, current, temperature, and other parameters. Although this product includes certain protection functions, operation beyond the limits may still cause irreversible damage.
④ Always disconnect the power promptly after use. If the battery remains connected, the ESC may unexpectedly drive the motor and create a safety hazard. Keeping the battery connected for a long time may also over-discharge the battery and may cause battery or ESC failure.

03 Product Introduction

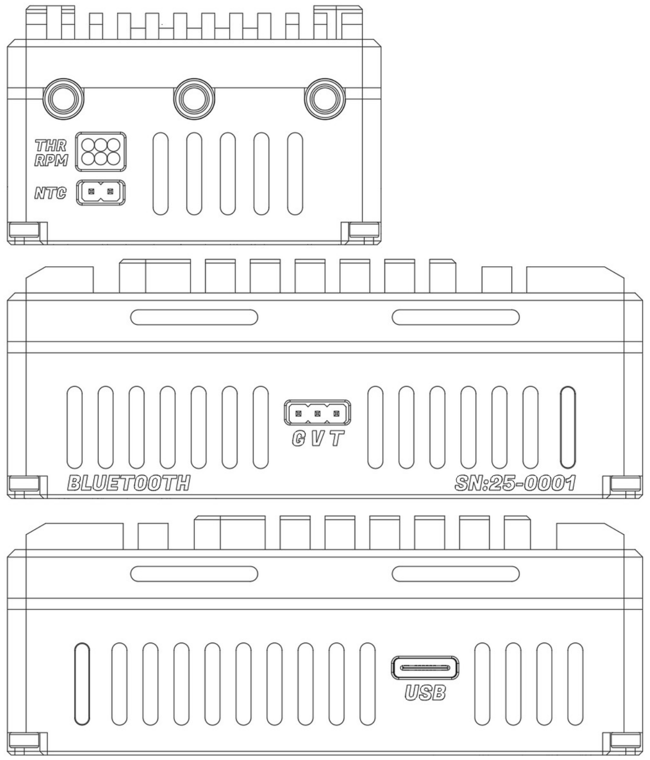
The FLYROTOR 280A HV ESC uses an industrial-grade, low-power microcontroller based on a 32-bit RISC-V core, designed specifically for high-performance motor control. The MCU features a dedicated hardware architecture, including an optimized stack area and fast interrupt mechanism, greatly improving interrupt response speed and ensuring smooth, precise control at a 120 MHz main frequency. It also integrates a high-precision 12-bit ADC, advanced timers, multiple operational amplifiers and comparators, and rich communication interfaces, greatly reducing external component requirements and simplifying system design and integration.
For the drive architecture, the ESC uses an independent pre-driver chip for each phase bridge arm. It directly drives a power array built from Infineon large-package TOLL, low-RDS(on), high-current MOSFETs. Independent drive channels ensure that each phase MOSFET receives sufficient and consistent drive signals under high-current conditions, significantly improving dynamic response and overall reliability.
In addition, this ESC has a built-in high-performance Active Freewheeling SBEC system. It supports continuous high-power output and integrates reverse-current protection and over-voltage protection circuits for both safety and practicality. Backup power sources such as a LiPo battery or rescue capacitor can be connected in parallel with ease, meeting the needs of diverse application scenarios.

04 Product Specifications

Model	FLYROTOR 280A HV
Drive Type	Brushless Motor
Drive Mode	Active Freewheeling, regenerative braking
Drive Frequency	10-24 kHz
Applicable Models	650-800 class electric helicopters, large electric fixed-wing aircraft
Input Voltage	6-14S LiPo battery
Operating Current	280/350 A (continuous/peak)
SBEC Voltage	Off, 7.5 V, 8.0 V, 8.5 V, 12 V
SBEC Current	10/35 A (continuous/peak)
Input Anti-spark	Supported
Input Wires	220 mm black and red 8 AWG silicone wires
Output Wires	130 mm black 10 AWG silicone wires
Throttle Cable	520 mm black/red/white 100-core parallel cable
RPM Cable	520 mm brown/red/yellow 100-core parallel cable
Motor Temperature Sensor	Supported
Parameter Setting Hardware	USB (built-in), Bluetooth (built-in), ROTORFLIGHT gyro
Parameter Setting Software	PC (Windows), command line (Windows/Linux/macOS), mobile (Android/iOS), transmitter (EdgeTX/ETHOS/Spirit)
LED Indicator	Colorful LED
Cooling Method	Aluminum alloy case, 4510 fan
Telemetry Protocol	FLYROTOR, KISS, S.BUS2, S.PORT, I.BUS2, CRSF, HoTT
Firmware Update	Offline and online update
Protection Functions	Low Voltage Protection, MOSFET Temperature Protection, Motor Temp Protection, Over Current Protection, Short Circuit Protection, Stall Protection, locked-rotor protection, throttle loss protection, SBEC short circuit protection, SBEC reverse-current protection, SBEC over-voltage protection, SBEC Fault Protection, battery consumed capacity protection
ESC Mode	ESC Governor, Linear Throttle, RF Gyro Governor
Dimensions	85 x 50 x 32 mm, 85 x 50 x 44.5 mm (with fan)
Weight	270 g (with wires), 290 g (with wires and fan)

Note: Actual product dimensions, wire length, weight, and other data may vary due to manufacturing tolerances. Please refer to the actual product.

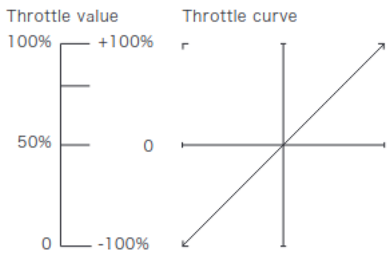
05 Communication Wires and Port Definitions



① ***THR***: Throttle signal cable (white, red, black), used as the ESC throttle signal input. Connect it to the receiver throttle channel or the flybarless system throttle channel, depending on the receiver type and flybarless system type. The white wire is the throttle signal wire, and the red and black wires are the SBEC wires.
② ***RPM***: RPM signal cable (yellow, red, black), used as the motor electrical RPM signal output. Connect it to the flybarless system RPM input channel. When using gyro governor mode, this RPM signal cable can provide the RPM signal input. The yellow wire is the RPM wire, and the red and black wires are the SBEC wires.
③ ***NTC***: External temperature sensor port. Connect an NTC temperature sensor with no polarity requirement. It can be used to detect ambient temperature or motor temperature.
④ ***GVT***: This port is a multifunction composite interface. The black (***G***) and red (***V***) wires supply power to the cooling fan. The purple (***T***) wire is used for ESC telemetry data output and transmitter forward programming input when using ROTORFLIGHT open-source or Spirit gyro systems.
⑤ USB: This port is a standard Type-C connector and supports reversible insertion. After connecting to a computer with a USB data cable, you can update firmware or adjust ESC parameters in the PC configuration software.

06 Throttle Range Calibration

The factory default ESC throttle range is 1100-1940 us. Recalibrate the throttle range when using the ESC for the first time or after switching to another transmitter.
Before calibrating the throttle range, set the throttle curve to Normal (45-degree diagonal line), and make sure the transmitter's highest throttle point corresponds to 100% throttle and the lowest throttle point corresponds to 0% throttle.



① Turn on the transmitter and move the throttle stick to the highest position (100%).
② Connect the battery to the ESC. The red LED stays on for 3 seconds, then the green LED turns on, and the motor emits the " ♪1-2-3" tone to indicate normal power-up.
③ The motor then emits two " ♪7-7" tones to confirm the highest throttle value.
④ Move the throttle stick to the lowest position (0%). The motor emits two " ♪1-1" tones to confirm the lowest throttle value.
⑤ Finally, the motor emits two " ♪1-5" tones, indicating that throttle range calibration is complete.

07 Normal Startup Procedure

① Turn on the transmitter and move the throttle stick to the lowest position (0%).
② Connect the battery to the ESC. The red LED stays on for 3 seconds, then the green LED turns on, and the motor emits the " ♪1-2-3" tone to indicate normal power-up.
③ The motor then emits two " ♪1-5" tones, indicating that the throttle signal is normal and ESC startup is complete.

08 ESC Parameter Settings and Information Viewing

1. Hardware
The ESC already integrates USB and Bluetooth modules, so no additional configurator is required.
2. Software
PC (Windows), command line (Windows/Linux/macOS), mobile (Android/iOS), transmitter (EdgeTX/ETHOS/Spirit).
3. Software and script download address
https://flydragonrc.com/download
4. Software and firmware updates
Both the configuration software and ESC firmware can be updated online.
5. ESC Info
ESC information will be read after the ESC is connected successfully, as shown below.
>ESC Type: Helicopter [280A]
>Serial Number: D202BED83A55A949
>IAP Version: 1.0.1
>Firmware Version: 1.1.0
>Hardware Version: 1.0
>Throttle Range: 1100-1940 us
>Total Motor Run Time: 00:10:15
>Total Consumed Capacity: 1.205 Ah
>Motor Run Count: 5

6. Log

The ESC has built-in non-volatile memory and supports recording the data from the most recent motor run, from successful motor start to motor stop. Starting the motor again will overwrite the previous log data. The recorded information is as follows:

Parameter Item	**Parameter Value**
Motor Run Time	Motor Run Time
Max Control Throttle	Maximum throttle signal value received by the ESC
Max Drive Throttle	Maximum throttle value used by the ESC to drive the motor
CPU Temperature	CPU temperature: minimum -> maximum
MOSFET Temperature	MOSFET temperature: minimum -> maximum
Motor Temperature Sensor	Motor temperature: minimum -> maximum
Battery Voltage	Battery voltage: maximum -> minimum
SBEC Voltage	SBEC voltage: maximum -> minimum

Maximum Current	Maximum bus current
Consumed Capacity	Accumulated consumed capacity
Maximum RPM	Maximum motor electrical RPM or head speed
Maximum Power	Maximum drive power
ESC Protection Events	Low Voltage Protection, MOSFET Temperature Protection, Motor Temp Protection, Throttle Loss Protection, Stall Protection, Short Circuit Protection, Over Current Protection, SBEC Fault Protection

09 Programmable Parameters and Descriptions

[General Parameters]

Parameter Item	Parameter Value	Default Value
ESC Mode	ESC Governor, Linear Throttle, RF Gyro Governor	ESC Governor
LiPo Cell Count	4-14S	6S
Low Voltage Limit	2.8-3.8 V	3.2 V
Temperature Limit	50-135°C	110°C
SBEC Output Voltage	Off, 7.5 V, 8.0 V, 8.5 V, 12 V	7.5 V
Electrical Angle	Auto, 1-10°	Auto
Motor Direction	Normal, Reverse	Normal
Starting Power	1-15	5
Response Speed	1-15	5
Beeper Volume	1-5	2
Current Gain	-20-0-20	0
Cooling Fan Mode	Temp Control, Always On, Always Off	Temp Control

1. ESC Mode

ESC Governor: For helicopters using ESC Governor. The motor starts when throttle is above 30%, using a smooth soft start. After soft start is complete, the ESC enters governor mode once the RPM stabilizes.

Linear Throttle: For fixed-wing aircraft or helicopters using an external governor device. The motor starts within the 5%-15% throttle range, using a smooth soft start. After soft start is complete, the ESC accelerates quickly to the current throttle value.

RF Gyro Governor: For helicopters using RF Gyro Governor. The startup logic and response speed are the same as Linear Throttle mode.

2. LiPo Cell Count

Set the battery cell count. The adjustable range is 4-14S. The setting must match the actual battery cell count in use.

3. Low Voltage Limit

Set the Low Voltage Limit. The adjustable range is 2.8-3.8 V per cell, with a step of 0.1 V. For example, when using a 6S battery, the protection voltage is the set value x 6. When the total battery voltage is lower than the protection value, the ESC will gradually reduce output power to 50%.

4. Temperature Limit

Set the MOSFET Temperature Protection threshold. The adjustable range is 50-135°C, with a step of 1°C. When the MOSFET temperature exceeds the set value, the ESC will gradually reduce output power to 50%.

5. SBEC Output Voltage

Set the output voltage of the built-in SBEC. The adjustable options are Off, 7.5 V, 8.0 V, 8.5 V, and 12 V.

6. Electrical Angle

Set the Electrical Angle for driving the motor. It can be selected automatically or manually. The adjustable range is 1-10°, with a step of 1°.

7. Motor Direction

Set the motor rotation direction. The options are Normal and Reverse.

8. Starting Power

Set the output power during motor startup. The adjustable range is 1-15. A higher value provides stronger starting power. A suitable value is one that allows smooth startup without tail kick.

9. Response Speed

Set the throttle response speed of the ESC in Linear Throttle and RF Gyro Governor modes. The adjustable range is 1-15. A higher value gives faster response.

10. Beeper Volume

Set the prompt beeper volume. The adjustable range is 1-5. A higher value gives louder volume.

11. Current Gain

Adjust the deviation between the consumed capacity recorded by the ESC and the actual capacity. The adjustable range is -20 to +20. If the capacity recorded by the ESC is lower than the capacity put back by the charger, increase the gain value; otherwise decrease it.

12. Cooling Fan Mode

Set the cooling fan mode. The options are Temp Control, Always On, and Always Off.

Temp Control: 2-second self-test at power-up. The fan turns on when temperature is above 45°C and turns off when temperature is below 40°C.

Always On: The fan runs continuously after the 2-second power-up self-test.

Always Off: The fan remains off at all times.

[Advanced Parameters]

Parameter Item	Parameter Value	Default Value
Soft Start Time	5-55 seconds	15 seconds
Autorotation Bailout Time	Off, 1-100 seconds	Off
Autorotation Bailout Acceleration	1-10	5
Governor P	0-100	45
Governor I	0-100	35
Active Freewheeling	Off, On	On
Drive Frequency	10-24 kHz	16 kHz
Maximum Motor ERPM	1000-1000000 ERPM	130000 ERPM

1. Soft Start Time

Set the duration of the soft start process in ESC Governor mode. The adjustable range is 5-55 seconds. A higher value makes the soft start process smoother.

2. Autorotation Bailout Time (fast restart)

This function is only effective in ESC Governor mode. After throttle is reduced from above 30% to the 20%-30% range, if throttle is raised above 30% again within the set Autorotation Bailout Time, the motor will skip the normal soft start process and quickly recover to the target RPM according to the set Autorotation Bailout Acceleration.

If throttle is below 20%, or if it stays in the 20%-30% range longer than the set Autorotation Bailout Time, this function becomes invalid, and the next throttle increase will perform the normal soft start process.

Safety warning: Do not test this function on the ground! During fast restart, the motor accelerates to the target RPM in a very short time, which may cause the model to spin up and surge suddenly, resulting in personal injury or equipment damage.

3. Autorotation Bailout Acceleration

This function is only effective in ESC Governor mode. Set the acceleration at which the motor recovers to the target RPM during autorotation bailout. The adjustable range is 1-10. A higher value accelerates faster.

4. Governor P

This function is only effective in ESC Governor mode. It sets the proportional compensation strength used by the ESC to maintain the target RPM. A higher value gives stronger correction when RPM deviation occurs. It must be adjusted together with Governor I.

5. Governor I

This function is only effective in ESC Governor mode. It sets the integral compensation strength used by the ESC for sustained RPM deviation. It adjusts the degree of RPM compensation. If the parameter is too high, over-compensation may occur, causing RPM oscillation; if it is too low, compensation will be insufficient and RPM recovery will be slow.

6. Active Freewheeling

Set the motor drive mode. In Linear Throttle or RF Gyro Governor mode, this function can be turned on or off. In ESC Governor mode, it is fixed to On. Enabling this function improves throttle linearity and drive efficiency.

7. Drive Frequency

Set the PWM frequency for driving the motor. The adjustable range is 10-24 kHz. The default value of 16 kHz is suitable for most motors, and it is recommended to keep the default unless there is a special requirement.

8. Maximum Motor ERPM

Set the maximum motor electrical RPM (ERPM). The calculation formula is as follows:

Maximum Motor ERPM = LiPo Cell Count x 4.2 V x Motor KV x (Motor Poles / 2)

Example: 12S battery, 520 KV motor, 10 poles

Maximum Motor ERPM = 12 x 4.2 x 520 x (10 / 2) = 131,040 ERPM

[Other Parameters]

Parameter Item	Parameter Value	Default Value
Throttle Protocol	PWM	PWM
Telemetry Protocol	FLYROTOR, KISS, S.BUS2, S.PORT, I.BUS2, CRSF, HoTT	FLYROTOR
LED Color	Custom, Off, Red, Green, Blue, Yellow, Magenta, Cyan, White, Orange, Gray, Maroon, Dark Green, Navy, Purple, Teal, Silver, Pink, Gold, Brown, Light Blue, Fluorescent Pink, Fluorescent Orange, Fluorescent Lime, Fluorescent Mint, Fluorescent Cyan, Fluorescent Purple, Fluorescent Bright Pink, Fluorescent Light Yellow, Fluorescent Aquamarine, Fluorescent Gold, Fluorescent Deep Pink, Fluorescent Neon Green, Fluorescent Orange Red	Green
Motor Temperature Sensor	On, Off	Off
Motor Temperature Limit	50-150°C	115°C
Consumed Capacity Limit	Off, 1-50000 mAh	Off

1. Throttle Protocol

Set the ESC throttle signal protocol. It is currently fixed to the PWM protocol, supporting a refresh rate of 50-480 Hz and a pulse width range of 950-2050 us. The factory default range is 1100-1940 us.

2. Telemetry Protocol

Set the telemetry data protocol output by the ESC. The following options are supported:

FLYROTOR: Bidirectional communication protocol defined by FLYROTOR ESCs. It supports ESC parameter settings and telemetry data return through transmitters using EdgeTX, ETHOS, or Spirit systems.

KISS, S.BUS2, S.PORT, I.BUS2, CRSF, HoTT: Can be connected directly to flight controllers or receivers that support the corresponding protocol.

Note: Hardware version 1.0 only supports the FLYROTOR, KISS, CRSF, and HoTT protocols.

3. LED Color

Set the LED color. Multiple preset colors and a custom mode are available. Red and Purple are already used by the system, so selecting another color for the user LED is recommended for easier identification.

LED Pattern

ESC Status	Display Mode
Idle	User LED on for 100 ms, off for 1000 ms
Settings Mode	User LED always on
Motor Running Normally	User LED always on
Throttle Range Calibration	Red LED always on
Autorotation Bailout Ready	Purple LED always on
Protection State	Red LED flashes. See <10 Protection Functions> for details.

4. Motor Temperature Sensor

Set whether to enable the external NTC temperature sensor function. When selecting On, make sure the sensor is correctly installed and connected to the specified ESC port.

5. Motor Temperature Limit

When the external temperature sensor is enabled and the probe is installed on the motor winding, the motor over-temperature protection temperature can be set. The adjustable range is 50-150°C, with a step of 1°C. When the detected temperature exceeds the set value, the ESC will gradually reduce output power to 50%.

6. Consumed Capacity Limit

Set the consumed capacity protection threshold accumulated by the ESC coulomb counter. The options are Off or manual selection, with an adjustable range of 1-50000 mAh and a step of 100 mAh. When the accumulated consumed capacity exceeds the set value, the ESC will gradually reduce output power to 50%.

[RPM Mode Parameters]

Parameter Item	Parameter Value	Default Value
RPM Mode	Electrical RPM, Head Speed	Electrical RPM
Motor Poles	2-100	10
Motor Gear	1-10000	11
Main Gear	1-10000	106

1. RPM Mode

Set the returned RPM data mode. Select Electrical RPM or Head Speed. The default is Electrical RPM. When Head Speed is selected, the ESC returns the calculated rotor head speed, and Motor Poles and gear counts must be set correctly.

2. Motor Poles

Set the motor pole count value, that is, the number of motor magnets. The adjustable range is 2-100. Enter the actual value. This data is only valid when RPM Mode is Head Speed.

3. Motor Gear

Set the motor gear tooth count. The adjustable range is 1-10000. Enter the actual value. This data is only valid when RPM Mode is Head Speed.

4. Main Gear

Set the main gear tooth count. The adjustable range is 1-10000. Enter the actual value. This data is only valid when RPM Mode is Head Speed.

[Melody Editor]

Custom startup melodies are supported. Up to 126 notes can be played. The melody format is RTTTL. Default melody:

cmaj:b=300,o=5,d=4:4c,4d,4e

10 Protection Functions

Protection Type	Alarm Signal Pattern	Beeper Count	Red LED Flash Count
Throttle Abnormal	*----	1 time	1 time
Motor Locked Rotor	*-*----	2 times	2 times
Low Voltage Protection	*-*-*----	3 times	3 times
MOSFET Temperature Protection	*-*-*-*----	4 times	4 times
Motor Temp Protection	*-*-*-*-*----	5 times	5 times
Over Current Protection	*-*-*-*-*-*----	6 times	6 times
Short Circuit Protection	*-*-*-*-*-*-*----	Silent	7 times
Stall Protection	*-*-*-*-*-*-*-*----	8 times	8 times
SBEC Fault Protection	*-*-*-*-*-*-*-*-*----	9 times	9 times

1. Symbol Description

*: Indicates motor beep (♪1 tone), red LED on, lasting 200 ms.

-: Indicates motor silent, red LED off, lasting 1000 ms.

----: Indicates one interval, lasting 1000 ms.

2. Soft Protection Mechanism

When Low Voltage Protection, MOSFET Temperature Protection, Motor Temp Protection, or Over Current Protection is triggered, the ESC will gradually reduce motor output power to 50%.

3. Hard Protection Mechanism

When throttle signal abnormality, Short Circuit Protection, motor Stall Protection, or SBEC Fault Protection is triggered, the ESC will immediately cut off power output.

4. Locked-Rotor Protection

During the motor startup stage, if 10 consecutive startup attempts fail, locked-rotor protection is triggered.

5. Protection Reset Conditions

Automatic recovery: After MOSFET Temperature Protection or Motor Temp Protection, the protection is automatically cleared when the temperature detected by the ESC drops below 70°C.

Manual recovery: All other protections can only be cleared by powering on again after the fault has been removed.

11 Special Functions

1. Extreme Low Voltage Protection

This is a secondary protection mechanism. After the ESC has limited power to 50% due to normal Low Voltage Protection, if the user still does not actively turn off power output and the battery voltage continues to drop below the extreme value of **2.4 V** per cell, the ESC will forcibly cut off power output to prevent complete battery damage caused by deep over-discharge.

12 After-sales Service Terms

1. Warranty Eligibility

Products that meet all of the following conditions are eligible for warranty service:

Warranty period:

① If valid proof of purchase cannot be provided, the product is covered by warranty for 90 days from the production date.

② If valid proof of purchase is provided, the product is covered by warranty for 365 days from the purchase date stated on the proof of purchase.

Fault nature: Our technical inspection confirms that the fault is caused by a quality issue of the product itself, including component defects or poor workmanship.

Warranty label: The warranty label attached to the product must remain intact and legible, with no missing or damaged parts.

Note: For products whose warranty period has expired and that have been returned for repair, an extended 60-day warranty service is provided from the date the repaired product is shipped back.

2. Warranty Exclusions

Products are no longer eligible for warranty if any of the following situations occur:

① The product has exceeded the warranty period specified in Section 1.

② The product has been disassembled by the user or its internal structure has been modified without authorization.

③ Damage caused by improper use by the user, such as water ingress or liquid corrosion, heavy impact or drop, improper configuration, and similar causes.

④ The product warranty label is missing or damaged.

3. Shipping Cost Responsibility

① **Mainland China:** For products that meet the warranty conditions, the customer is responsible for shipping costs to our company, and our company is responsible for return shipping costs after repair.

② **Outside mainland China:** Regardless of whether the product is within the warranty period, round-trip shipping costs are borne by the customer. We recommend contacting your local dealer first to arrange consolidated returns, reducing courier and bank handling fees.

3. **Software upgrade service:** If our company is entrusted to perform a software upgrade, the resulting round-trip shipping costs are borne by the customer.

Note: When sending repair items, please use a reputable courier service. Overseas customers are advised to use FedEx or UPS. After shipment, please proactively inform our after-sales department so that we can receive and process the item promptly.

4. Seven-day Replacement Service

If the product has no obvious appearance damage and our technical personnel confirm a quality issue within 7 days from the purchase date, we will provide replacement service.

5. Service Process Guide

Contact after-sales -> Return for repair -> Inspection and evaluation -> Customer confirmation -> Repair processing -> Product return shipment

① When you confirm that the product is abnormal or damaged, contact our after-sales service personnel by phone, email, WeChat, or other methods.

② After-sales service personnel will first help you determine whether the issue is caused by operation.

③ If return-to-factory repair is required, please include a note in the package stating the contact person's name, shipping address, phone number, fault description, and repair request.

④ After receiving the product, we will inspect it. If repair fees are required, we will explain the fee details and payment method to you by phone, email, WeChat, or other methods, and repair will proceed only after you confirm and agree.

⑤ After the product is repaired, after-sales personnel will confirm the shipping information with you. After receiving the relevant payment, we will arrange return shipment as soon as possible. Under normal circumstances, we will complete the repair within 3 working days from confirmation of the repair plan.